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PROJECT TITLE: DESIGN AND IMPLEMENTATION OF A MOBILE APP FOR COLLECTION OF USER EXPERIENCE DATA FROM MOBILE NETWORK SUBSCRIBERS.

TASK 4: SYSTEM MODELLING AND DESIGN

GROUP 2

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1. INTRODUCTION

This report presents a comprehensive system modelling and design for a mobile application engineered to collect Quality of Experience (QoE) data in real time from mobile network subscribers. The application is purpose-built for the Cameroonian mobile network landscape – an environment characterized by persistent network volatility, inconsistent service delivery, and recurring connectivity outages that directly impact the daily digital experience of millions of subscribers.

As mobile data consumption continues to grow across Cameroon, the absence of structured, subscriber-side quality monitoring leaves network operators with limited visibility into the true end-user experience. This application addresses that gap by enabling continuous, automated collection of network performance metrics directly from the subscriber's device, providing operators with ground-truth data that traditional network-side monitoring cannot capture.

The system design is articulated through six complementary UML and structured-analysis diagrams, each serving a distinct architectural purpose. A Context Diagram establishes the system's external boundaries and key stakeholder interactions. A Data Flow Diagram (DFD) traces the movement and transformation of data across system processes. A Use Case Diagram defines the functional scope from the perspective of each actor. Sequence Diagrams illustrate the step-by-step message flows between system components during critical operations. A Class Diagram captures the underlying data structure, object relationships, and system logic. Finally, a Deployment Diagram maps the physical distribution of software components across hardware infrastructure.

Together, these diagrams form a unified architectural blueprint, guiding the system from initial user interaction through data collection, local storage, server-side processing, and ultimately to operator-facing analytics ensuring that every layer of the application is clearly defined, logically consistent, and ready for implementation.

2. CONTEXT DIAGRAM

2.1 Overview

The Context Diagram (also referred to as a Level-0 Data Flow Diagram) defines the boundary of the QoE Mobile Application system and its interactions with external entities. It presents the system as a single high-level component surrounded by the actors and external systems it communicates with.

2.2 System Components and Actors

The context diagram identifies the following components:

- **QoE Mobile App** The central system under design. Acts as the primary collector and relay of QoE data.
- **Mobile Subscriber** The end-user who provides subjective feedback (satisfaction ratings) and whose device passively generates objective network metrics.
- **Network Operators** Administrators and analysts who consume aggregated QoE data via the dashboard.
- **External APIs (GPS / Network Info)** Third-party Android system APIs that supply signal strength (dBm), network type (3G/4G/5G), and GPS coordinates.
- **Backend Server** A cloud-hosted REST API server that receives uploaded metric batches and serves processed data.
- **Local DB (SQLite)** A local on-device database used for offline buffering of collected data.
- **QoE Dashboard (Web UI)** A web-based interface through which network operators view analytics and generate reports.

2.3 Key Interactions

From	To	Data / Interaction
Mobile Subscriber	QoE Mobile App	Uses (provides feedback, grants permissions)
QoE Mobile App	External APIs	API Request for signal/GPS data
QoE Mobile App	Local DB (SQLite)	Persist records locally
QoE Mobile App	Backend Server	HTTP upload of metric batches
Backend Server	QoE Dashboard	Serves aggregated data via HTTP
Network Operators	QoE Dashboard	Views analytics and reports

3. DATA FLOW DIAGRAM (DFD)

3.1 Overview

Two levels of DFD have been modelled: a Level-0 (Context) model and a Level-1 decomposition. The Level-1 DFD breaks the QoE Mobile App process into four discrete subprocesses and two data stores, illustrating the internal flow of data within the system.

3.2 Level-0 DFD Processes and Data Flows

The top-level DFD identifies five main processes:

- Mobile Subscriber Sends satisfaction ratings; receives feedback prompts.
- QoE Mobile App Central process hub; issues API requests, generates QoE metric reports, persists metric records.
- External APIs (Cell Tower / GPS) Returns signal strength and location data.
- Backend Server (Cloud API) Receives uploaded data and returns Sync ACK confirmations.
- Network Operator (Admin/Analyst) Provides configuration and threshold inputs.

3.3 Level-1 DFD Decomposed Processes

The Level-1 model introduces the following sub-processes and data stores:

Process ID	Process Name	Description
P1	Metric Collector	Receives Signal/GPS data from External APIs; writes metric records to DS1.
P2	Feedback Capture	Prompts the subscriber; captures rating/consent; writes feedback records to DS1.
P3	User Settings	Reads stored metrics; generates QoE reports; handles operator config inputs.
P4	Data Sync	Reads pending records from DS1; queues batches in DS2; uploads to Backend Server; marks records as synced on ACK.

Data Stores:

- DS1 Local SQLite DB: Central on-device repository for all raw metric and feedback records.
- DS2 Upload Queue: Temporary staging store for batches pending synchronisation with the backend.

4. USE CASE DIAGRAM

4.1 System Boundary

The Use Case Diagram defines the boundary of the QoE Mobile App System and enumerates all the functional interactions it supports for its two primary actor types: Mobile Subscribers and Network Operators.

4.2 Actors

- Mobile Subscriber A mobile phone user who installs the application and participates in data collection.
- Network Operator An administrative user (analyst or engineer) who accesses the system's analytics capabilities.

4.3 Use Cases

Use Case	Actor(s)	Description
Submit Satisfaction Rating	Mobile Subscriber	User rates their current network experience on defined dimensions (satisfaction, speed, usability).
View Privacy Notice	Mobile Subscriber	User reads the data collection privacy statement.
Configure Prompt Frequency	Mobile Subscriber	User sets how often the app requests feedback (minimum once per day).
Grant or Revoke Permissions	Mobile Subscriber	User controls Android permissions for location, network access, etc.
Request Data Deletion	Mobile Subscriber	User triggers deletion of all locally and remotely stored personal records.
Collect Network Metrics	System (Background)	App autonomously collects signal strength, network type, GPS, latency, etc.
Sync Data to Server	System (Background)	App batches and uploads pending records to the backend.
View QoE Dashboard	Network Operator	Operator logs in and views aggregated charts, heatmaps, and time-series data.
Configure Alert Thresholds	Network Operator	Operator sets minimum acceptable metric values (e.g. dBm < -100).
Export Metrics Report	Network Operator	Operator generates and downloads a CSV or PDF report of collected metrics.

5. SEQUENCE DIAGRAMS

Ten sequence diagrams have been modelled, each corresponding to one use case identified in the Use Case Diagram. They describe the time-ordered message exchanges between system lifelines for each scenario.

5.1 Submit Satisfaction Rating

Lifelines: Subscriber, Feedback Service, Metric Collector, Local DB

- Feedback Service prompts the subscriber with a feedback dialog.
- Subscriber submits their rating.
- Metric Collector captures a real-time snapshot: dBm signal, GPS coordinates, and timestamp.
- The metric and rating records are persisted to Local DB via an INSERT operation.
- Local DB returns ACK; a confirmation is displayed to the subscriber.

5.2 Configure Prompt Frequency

Lifelines: Subscriber, Settings Service, Local DB

- Subscriber opens the settings screen.
- Settings Service reads the current prompt interval from Local DB and displays it.
- Subscriber sets a new interval (e.g., every 6 hours); minimum enforced is once per day.
- Updated value is written to Local DB and ACK is returned; confirmation is shown.

5.3 Grant / Revoke Permissions

Lifelines: Subscriber, QoE App, Android OS, Local DB

- App requests required Android permissions on launch or when needed.
- Android OS displays the permission dialog to the subscriber.
- Subscriber grants or denies; Android OS returns the permission result to the app.
- App saves the consent decision to Local DB and enables or disables the relevant data collectors.

5.4 View Privacy Notice

Lifelines: Subscriber, QoE App

- Subscriber opens the privacy notice from the app menu.
- QoE App renders the full privacy statement.
- Subscriber acknowledges and closes the screen.

5.5 Request Data Deletion

Lifelines: Subscriber, QoE App, Local DB, Backend Server

- Subscriber initiates a deletion request.
- App shows a confirmation dialog; subscriber confirms.
- App issues DELETE ALL RECORDS to Local DB; receives local ACK.
- App sends DELETE /data/{device_id} to Backend Server; receives HTTP 200 OK.
- Deletion confirmed to subscriber.

5.6 Collect Network Metrics (Background)

Lifelines: Background Timer, Metric Collector, External APIs, Local DB

- Background timer fires onTimer() every 5 minutes.
- Metric Collector calls External APIs: GET signal strength (dBm), GET network type (3G/4G/5G), GET GPS coordinates (latitudes, longitudes, accuracy).
- All returned values are inserted as a single metric record into Local DB; row_id ACK returned.

5.7 Sync Data to Server

Lifelines: Sync Service, Local DB, Upload Queue (DS2), Backend Server

- Sync Service reads all pending (unsynced) records from Local DB.
- Records are enqueued in DS2 as a batch.
- Batch is transmitted via POST /metrics (JSON) to Backend Server.
- Backend returns 200 OK + sync_id; records are marked as uploaded in Local DB.

5.8 View QoE Dashboard

Lifelines: Network Operator, Dashboard UI, Backend Server

- Operator logs in and opens the dashboard.
- Dashboard UI requests GET /metrics?range=7d from Backend Server.
- Backend returns aggregated metrics JSON; charts and heatmaps are rendered.
- Operator applies a regional filter; Dashboard issues GET /metrics?region=x; filtered results update the display.

5.9 Configure Alert Thresholds

Lifelines: Network Operator, Backend Server, QoE App

- Operator posts new threshold configuration: POST /thresholds {dBm: -100}.
- Backend validates and stores the configuration; returns 200 OK.

- Backend pushes the updated configuration to active QoE App instances; app returns ACK.

5.10 Export Metrics Report

Lifelines: Network Operator, Backend Server, Export Service

- Operator requests an export in CSV or PDF format.
- Backend aggregates the relevant dataset and passes it to the Export Service.
- Export Service formats the report and returns a download link or file to the operator.

6. CLASS DIAGRAM

6.1 Overview

The Class Diagram models the static structure of the QoE system's software objects, their attributes, operations, and the associations between them. Six core classes are identified.

6.2 Classes, Attributes, and Operations

Class	Key Attributes	Key Operations	Associations
Subscriber	subscriberId, name, phoneNumber, location	login(), logout(), updateProfile()	uses MobileApp
MobileApp	appVersion, isConnected, collectionInterval	collectData(), sendData(), syncToServer(), saveLocally()	Collects QoEMeasurement; Persists to LocalDatabase; HTTP to BackendServer
QoEMeasurement	measurementId, timestamp, signalStrength, latency, packetLoss, downloadSpeed, uploadSpeed, gpsLocation	recordMeasurement(), getMeasurementData(), calculateScore()	Collected by MobileApp
LocalDatabase	dbPath, dbVersion, lastSyncTime	saveRecord(), getRecords(), syncWithServer(), clearOldRecords()	Syncs to BackendServer; used by MobileApp
BackendServer	serverUrl, apiVersion, isOnline	receiveData(), processData(), storeData(), serveData()	Serves Dashboard; receives from MobileApp
Dashboard	dashboardUrl, refreshRate, activeUsers	displayData(), generateReport(), filterData()	Served by BackendServer

6.3 Key Relationships

- Subscriber uses MobileApp association representing the subscriber's interaction with the app.

- MobileApp collects QoEMeasurement association through the collectData() operation.
- MobileApp persists to LocalDatabase dependency (Persist) for offline storage.
- MobileApp communicates with BackendServer dependency (HTTP) for data upload.
- LocalDatabase syncs to BackendServer dependency ensuring data consistency.
- BackendServer serves Dashboard association enabling operator analytics.

7. DEPLOYMENT DIAGRAM

7.1 Overview

The Deployment Diagram shows the physical distribution of the system's software artefacts across hardware nodes and the communication links between those nodes.

7.2 Hardware Nodes

Node	Deployed Artefacts	Description
Mobile Device	QoE Mobile App, Local DB (SQLite)	Android smartphone used by the subscriber. Hosts the app and its local buffer database.
Application Server	Backend Server	Cloud-hosted server that receives metric uploads, processes data, and exposes the REST API. Connects to Database Server via JDBC/SQL.
Database Server	QoE Database	Centralised relational database (e.g., PostgreSQL/MySQL) storing all processed QoE records.
Web Server	QoE Dashboard (Web UI)	Serves the operator-facing web dashboard. Operators access it via browser.

7.3 Communication Links

- Mobile Device → Application Server: HTTP/REST (QoE Mobile App uploads metric batches to Backend Server).
- Application Server → Database Server: JDBC/SQL (Backend Server reads/writes to QoE Database).
- Application Server → Web Server: HTTP (Application Server serves data to the Dashboard).
- Mobile Subscriber → Mobile Device: Direct physical use of the app.
- Network Operator → Web Server: Browser-based access to the Dashboard.

8. SUMMARY AND DESIGN RATIONALE

8.1 Architectural Overview

The QoE data collection system follows a three-tier client-server architecture: a mobile client tier (Android app + SQLite), an application server tier (REST API backend), and a presentation tier (web dashboard). This separation of concerns ensures that the mobile app remains lightweight and functional in offline conditions, while centralizing data storage and analytics on the server.

8.2 Design Decisions

- Offline-first with sync: The use of a local SQLite database (DS1) and an upload queue (DS2) ensures that metric collection continues uninterrupted even when network connectivity is unavailable.
- Background collection every 5 minutes: As shown in the sequence diagram for metric collection, a background timer drives automated data capture without requiring user interaction.
- Privacy by design: Dedicated use cases and sequence flows for permission management, privacy notice display, and data deletion reflect compliance with user consent principles.
- Role-based access: The system serves two distinct actor roles mobile subscribers and network operators with entirely separate interaction paths and interfaces.
- RESTful communication: All mobile-to-server and server-to-dashboard communication uses HTTP/REST, ensuring scalability and ease of integration.

8.3 Diagrams Summary Table

Diagram	Type	Primary Purpose
Context Diagram	Structural	Defines system boundary and external actors/systems.
Data Flow Diagram	Behavioural	Shows data flows between processes and data stores at L0 and L1.
Use Case Diagram	Behavioural	Enumerates all functional interactions for each actor.

Sequence Diagrams (x10)	Behavioural	Details time-ordered message sequences for each use case.
Class Diagram	Structural	Models' system objects, their attributes, operations, and relationships.
Deployment Diagram	Physical	Maps software artefacts to physical hardware nodes and network links.